

Math Class 10 Federal Board

Chapter 01 Complex Numbers

Miscellaneous Exercise

MISCELLANEOUS EXERCISE-1

1. Tick the correct option.

i. $\sqrt{-1}$ is equal to:

a. 1

b. -1

c. i

d. $-i$

ii. If $x < 0$, then $\sqrt{x}$ is:

a. Real

b. Complex

c. Irrational

d. Rational

iii. Conjugate of $\sqrt{x} - i\sqrt{y}$ is:

a. $\sqrt{x} + i\sqrt{y}$

b. $x - iy$

c. $x - y$

d. $x + iy$

$$\begin{array}{c} \text{---ve} \quad | \quad \text{+ve} \\ \text{---} \quad 0 \quad \text{---} \end{array} \rightarrow \sqrt{-0.2}$$

$$\frac{0.2}{10} = \frac{1}{5} \times \frac{\sqrt{5}}{\sqrt{5}}$$

$$\frac{\sqrt{5}}{5} i$$

$$\sqrt{0.2} i$$

$$\bar{z} = x - iy \Rightarrow z\bar{z} = x^2 + y^2$$

iv. If $z = x + iy$, then $z\bar{z}$ is:

a. Imaginary

c. Non-negative number

b. Complex

d. Negative number

v. $\sqrt{-25} + \sqrt[3]{8}$ is equal to:

a. $-5 + \sqrt{8}$

b. $2 + 5i$

c. $-5 + 2i$

d. $2\sqrt{2} + 5i$

vi. $1 + (-i)^9 = ?$

a. $1 + i$

b. $1 + \sqrt{-1}$

c. $1 - i$

d. $-i$

vii. $\frac{2}{1-i} = ?$

a. $\frac{1+i}{2}$

b. $\frac{(1+i)^2}{2}$

c. $1 - i$

d. $1 + i$

a. $\frac{1+i}{2}$

b. $\frac{(1+i)^2}{2}$

c. $1-i$

d. $1+i$

v) $\sqrt{-25} + \sqrt[3]{8}$

$= \sqrt{25 \times -1} + (8)^{\frac{1}{3}}$

$= \sqrt{25} \times \sqrt{-1} + (2^3)^{\frac{1}{3}}$

$= 5i + 2 = \boxed{2+5i}$

v) $1 + (-i)^9$

$= 1 - i^9$

$= 1 - i \cdot i^8$

$= 1 - i (i^2)^4$

$= 1 - i (-1)^4$

$= \boxed{-i} \checkmark$

$(\because (-i)^1 (-1 \times i)^9 = (-1) \times i^9)$

$\boxed{i^2 = -1}$

$8 = 2 \times 4$

viii. $(-xi)^{19} = ?$

a. $-x^{19}i$

b. $x^{19}i$

c. $-i^{19}$

d. $-x^{19}$

a. $-x^{19}i$ b. $x^{19}i$ c. $-i$

$$\begin{aligned}
 (-xi)^{19} &= (-1)^{19} \cdot x^{19} \cdot i^{19} & 19 = 18+1 \\
 &= -x^{19} \cdot i^{18} \\
 &= -x^{19} \cdot i \cdot (i^2)^9 \\
 &= -x^{19} \cdot i \cdot (-1)^9 \\
 &= x^{19} \cdot i
 \end{aligned}$$

ix. If $z = 3 + 4i$, then $|z|^2$ is:

a. 5

b. $\sqrt{5}$

c. 25

d. 16

x. The solution of $x^2 + 4 = 0$ is:a. $2i$ b. $-2i$ c. ± 2 d. $\pm 2i$

$$|z| = \sqrt{9 + 16} = \sqrt{25}$$

$$|z|^2 = (\sqrt{25})^2 = 25$$

$$x^2 + 4 = 0$$

$$i^2 = -1$$

$$x^2 = -4$$

$$x^2 = 4i^2$$

$$\sqrt{x^2} = \pm \sqrt{(2i)^2}$$

$$x = \pm 2i$$

2. Simplify:

a. $(-2+4i)-(8-5i)$

b. $(-3+4i)+(-7i+4)$

$$\begin{array}{l|l}
 \begin{aligned}
 &= -2+4i-8+5i \\
 &= (-8-2)+(4i+5i) \\
 &= -10+9i \quad \text{Ans}
 \end{aligned}
 &
 \begin{aligned}
 &= -3+4+4i-7i \\
 &= 1-3i \quad \text{Ans}
 \end{aligned}
 \end{array}$$

3. Find the product:

a. $(x+iy)(2+3i)$

b. $(-3+6i)(-6+3i)$

a) $(x+iy)(2+3i)$

$= (x+iy)(2+3i)$

$$\begin{aligned}
 b) &= -3(-6+3i)+6i(-6+3i) \\
 &= 18-9i-36i+18i^2 \\
 &= 18-18-45i \\
 &= -45i \quad \text{Ans}
 \end{aligned}$$

$$\begin{aligned}
 &= x(2+3i) + iy(2+3i) \\
 &= 2x + \underline{3xi + 2yi} + 3yi^2 \\
 &= (2x - 3y) + (3x + 2y)i \quad \text{A}
 \end{aligned}$$

4. Write in the form of $a+bi$, then find its conjugate:

a. $3\sqrt{2} - \sqrt{-7}$

b. $\sqrt{-2}$

Let $z_1 =$
a) $3\sqrt{2} - \sqrt{7}i$

$$\bar{z}_1 = 3\sqrt{2} + \sqrt{7}i \quad \text{A}$$

Let $z_2 = \sqrt{-2}$

$$\begin{aligned}
 z_2 &= \sqrt{2}i \\
 \boxed{\bar{z}_2 &= -\sqrt{2}i} \quad \text{A}
 \end{aligned}$$

5. Find $z \cdot \bar{z}$:

a. $z = \frac{-1}{2} + i$

a) $\bar{z} = -\frac{1}{2} - i$

$$z \bar{z} = \left(-\frac{1}{2} + i\right) \left(-\frac{1}{2} - i\right)$$

$$= \left(-\frac{1}{2}\right)^2 - (i)^2$$

b. $z = 14 - 7i$

b) $\bar{z} = 14 + 7i$

$$\begin{aligned}
 z \bar{z} &= 14^2 + 7^2 \\
 &= 196 + 49 \\
 &= 245 \quad \text{A}
 \end{aligned}$$

$$= \left(-\frac{1}{2}\right) - (2)$$

$$= \frac{1}{4} - 2i^2$$

$$= \frac{1}{4} - (-1)$$

$$= \frac{1}{4} + 1$$

$$= \frac{5}{4} = 1\frac{1}{4}$$

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 $4\sqrt{\frac{5}{4}}$
 $\frac{4}{1}$

6. Simplify:

a. $\frac{-3-i}{-3+i}$

$$\text{Let } z = \frac{-3-i}{-3+i} \times \frac{-3-i}{-3-i}$$

$$= \frac{(-3-i)^2}{(-3)^2 + (1)^2}$$

$$= \frac{(-3)^2 + (-i)^2 + 2(-3)(-i)}{9+1}$$

$$= -1 + 6i$$

$$z =$$

b. $\frac{1+3i}{i\sqrt{5}}$

$$z = \frac{1+3i}{i\sqrt{5}} \times \frac{\sqrt{5}i}{\sqrt{5}i}$$

$$= \frac{\sqrt{5}i + 3\sqrt{5}i^2}{(\sqrt{5})^2 \cdot i^2}$$

$$= \frac{-3\sqrt{5} + \sqrt{5}i}{-5}$$

$$= \frac{3\sqrt{5}}{5} - \frac{\sqrt{5}i}{5}$$

$$= \frac{3}{\sqrt{5}} - \frac{1}{\sqrt{5}}i$$

$$= \frac{9 - 1 + 6i}{10}$$

$$= \frac{8 + 6i}{10}$$

$$= \frac{8}{10} + \frac{6i}{10}$$

$$= \boxed{\frac{4}{5} + \frac{3i}{5}} \quad \text{A}$$

$$\sqrt[7]{\sqrt{5} \cdot \sqrt{5}} \quad \checkmark$$

$$\frac{(5)^{1/2}}{5^1} = \frac{1}{5^{1-1/2}}$$

7. Factorize:

a. $2x^2 + 18$

$$\begin{aligned} a) &= 2(x^2 + 9) \\ &= 2(x^2 - (-9)) \\ &= 2(x^2 - 9i^2) \\ &= 2((x)^2 - (3i)^2) \end{aligned}$$

b. $-x^2 - 25y^4$

$$\begin{aligned} b) &= -(x^2 + 25y^4) \\ &= -(x^2 - (-25y^4)) \\ &= -(x^2 - 25y^4 i^2) \\ &= -(x^2 - (5y^2 i)^2) \\ &= -(x - 5y^2 i)(x + 5y^2 i) \end{aligned} \quad \text{A}_2$$

$$= 2(x-3i)(x+3i)$$

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8. Solve:

a. $3x^2 + 15 = 0$

$$\Rightarrow 3(x^2 + 5) = 0$$

$$\Rightarrow x^2 + 5 = 0$$

$$\Rightarrow x^2 = -5$$

$$x^2 = 5i^2$$

$$\sqrt{(x)^2} = \pm \sqrt{(5i)^2}$$

$$\boxed{x = \pm \sqrt{5}i}$$

$$x = \{-\sqrt{5}i, +\sqrt{5}i\}$$

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b. $6y^2 + 36 = 0$

$$\Rightarrow 6(y^2 + 6) = 0$$

$$\Rightarrow y^2 + 6 = 0$$

$$\Rightarrow y^2 - (-6) = 0$$

$$y^2 - 6i^2 = 0$$

$$(y)^2 - (\sqrt{6}i)^2 = 0$$

$$(y - \sqrt{6}i)(y + \sqrt{6}i) = 0$$

$$\Downarrow$$

$$y = \sqrt{6}i, \quad y = -\sqrt{6}i$$

$$S.S. = \{\sqrt{6}i, -\sqrt{6}i\}$$

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